

MAGNET-FREE EV TECH Was Inevitable, But Now It Seems Urgent

Amid global concerns over rare-earth supply chains and copper price volatility, the UK's AEM and Sterling Tools recently held a virtual session unveiling magnet and copper-free motors—proven in Europe and now being localised for India.

Sterling GTake
E-Mobility (SGEM)
and Advanced
Electric Machines
(AEM)

Rare-earth elements, essential for modern technologies such as electric vehicles (EVs), wind turbines, and defence systems, have been the focal point of global economic and geopolitical tensions for some time.

As of mid-2025, China accounts for approximately 60% of global rare-earth mining and over 85% of the refining capacity. It also dominates the downstream supply chain, producing more than 90% of rare-earth magnets worldwide, according to a report by *Wion*.

That dominance has come under renewed scrutiny. Starting in late 2023, China began tightening export controls on critical rare-earth elements like terbium and dysprosium, which are key to high-performance electric motors. By April 2024, Beijing had mandated that exporters acquire special licences, citing national security concerns and potential military applications. These restrictions followed earlier curbs on gallium

and germanium, used in semiconductors and solar panels.

While China later introduced 'green lanes' to simplify rare-earth exports for select trading partners, such as the EU, the European Parliament responded strongly, passing a motion condemning China's move. The non-binding but influential motion, adopted ahead of the upcoming EU-China summit, described China's actions as coercive and unjustified. It urged the bloc to resist strategic pressure and avoid making trade concessions tied to resource access.

India, too, is feeling the pressure. Despite holding the world's third-largest reserves of rare-earth elements (REEs), the country has historically struggled to meet production targets due to regulatory, technical, and environmental hurdles. But reportedly, that is changing. According to *The Economic Times*, the recently launched National Critical Mineral Mission (NCMM) aims to unlock India's reserves through

domestic exploration, public-private partnerships, and strategic alliances—particularly with resource-rich countries like Australia.

Public sector units, such as IREL, are expanding their operations, while private companies like Sona Comstar, Vedanta, and

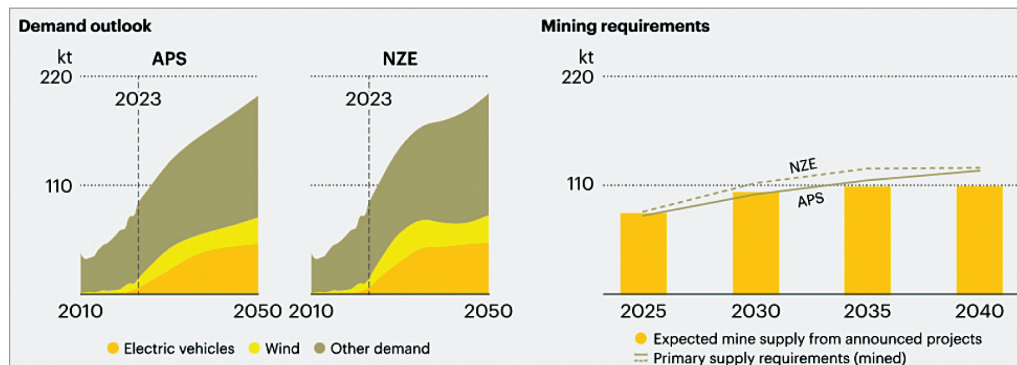


Fig. 1: IEA projects rapid growth in rare earth demand leading to shortages in the early 2030s (Courtesy: AEM, Sterling GTake)